

COSMOS LEDGER LABS

INFRASTRUCTURE SOFTWARE FOR AUTOMATED
DIGITAL ASSET SYSTEMS

ARCHITECTURE DIAGRAM

★ ECOSYSTEM PLATFORM HAS SECURED EXTERNAL FUNDING ★

PLATFORM OVERVIEW



COSMOS LEDGER LABS is designed as a secure workflow automation platform for digital asset operations.



The system is structured around workflow orchestration, operational security, and scalable execution infrastructure.

HIGH-LEVEL ARCHITECTURE DIAGRAM



High-Level Architecture

Admin Dashboard
(Web App / Control UI)

Workflow Builder
Create / Configure
Operational Workflows

Workflow Orchestrator
Execution Scheduling Engine
Validation & Routing Layer

Approval Engine
Multi-Step Approval
Permission Controls

Data & Logging
Workflow History
Audit Records
Operational Analytics

Security Layer
Permissions
Audit Logs
Atomic Rollback Protection
Execution Verification

Monitoring Layer
Execution Status Tracking
Failure Detection
Retry & Rollback Validation

Transaction Execution—Solana Smart Contract Layer
PDA/CPI/Transaction Logic

Token Issuance	Vesting Engine	Distribution
Smart Contracts	Lock Schedules	Execution Engine
Mint Operations	Release Logic	Batch Transfers

CORE SYSTEM COMPONENTS

01 Admin Dashboard

Primary operational interface for:
Workflow management
Distribution management
Vesting setup
Monitoring
Approval operations
Execution tracking

02 Workflow Builder

Responsible for:
Creating programmable
operational workflows
Defining execution rules
Configuring vesting schedules
Configuring distribution logic
Trigger management

3. Approval Engine

Responsible for:



Multi-step approvals



Permission validation



Role-based access control



Operational authorization

3A. Wallet & Permission Infrastructure

The platform is designed to support secure wallet coordination and permission-based operational execution.

Core infrastructure concepts may include:



4. Workflow Orchestrator

Core operational layer responsible for:

- Workflow scheduling
- Execution coordination
- Validation routing
- Trigger execution
- Transaction preparation

4A. Execution Flow Lifecycle

- User Creates Workflow
- Approval Validation
- Execution Queue
- Transaction Builder
- On-Chain Validation
- Transaction Execution
- Execution Monitoring
- Audit Logging
- Rollback Recovery

CORE SYSTEM COMPONENTS

5. Token Issuance Layer

Handles:

Token minting

Supply configuration

Issuance execution

Mint authority

management

6. Vesting Engine

Handles:

Token lock schedules

Linear vesting

Timed releases

Allocation tracking

7. Distribution Engine

Handles:

**Automated
distributions**

Batch transfers

Treasury operations

**Scheduled token
transfers**

8. Solana Smart Contract Layer

Infrastructure execution layer responsible for:

- PDA logic
- CPI execution
- Transaction validation
- On-chain execution
- Smart contract coordination

8A. Backend Infrastructure Layer

The backend infrastructure layer is designed to coordinate workflow execution, operational monitoring, and scalable system orchestration.

Core components may include:

- API Gateway
- Execution Queue
- Database Layer
- Event Processing System
- Monitoring Services
- Retry Coordination
- Notification Services
- Operational Logging

9. Monitoring Layer

Responsible for:

- Execution tracking
- Failure monitoring
- Status reporting
- Retry logic
- Rollback verification

8. Solana Smart Contract Layer

The platform is designed to support operational recovery and execution validation during workflow failures.

Core recovery concepts may include:

- Execution Failure Detection
- Retry Coordination
- Partial Execution Prevention
- Rollback Recovery
- Operational Recovery Validation
- Transaction Integrity Monitoring

The recovery system is intended to improve operational reliability and execution safety across automated workflows.

9B. Future Quantum-Resistant Security Considerations

The platform architecture is designed with future security extensibility in mind, including the potential integration of quantum-resistant security models and next-generation cryptographic standards as the digital asset infrastructure ecosystem evolves.

Future areas of exploration may include:

Advanced Cryptographic
Validation

Enhanced Operational Security
Frameworks

Quantum-Resistant Signature
Schemes

Long-Term Infrastructure Resilience

These considerations are part of the platform's long-term infrastructure vision and are not part of the current lightweight MVP implementation.

Infrastructure Positioning

COSMOS LEDGER LABS is positioned as:
Secure Workflow Automation
Infrastructure for Digital Asset Operations.
The platform is designed to provide operational coordination, programmable execution workflows, and scalable infrastructure tooling for Web3 ecosystems.

Future Expansion Potential

Potential future expansion may include:
API integrations
Multi-chain support
Enterprise workflow tooling
Advanced orchestration systems
Ecosystem integrations
External treasury integrations



Security Layer

Responsible for:
Permissions
Audit logs
Atomic rollback protection
Operational verification
Security controls

Security Design Goals

Prevent double execution
Prevent replay attacks
Improve operational reliability
Support audit-ready architecture
Enable scalable operational workflows
Improve execution visibility



THANK YOU

info@cosmosledgerlabs.com | cosmosledgerlabs.com